NEXROUTE: NEXT- GENERATION AUTONOMOUS DELIVERY SYSTEM

A line-following robot for secure indoor logistics — an autonomous robot that follows floor tracks to deliver confidential documents and materials in offices, universities, and corporate environments safely and securely.

Team Name: STATUS 200 OK

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Robotics Lab

Abstract

Our project aims to develop a smart indoor service robot that can assist people by carrying and delivering small items within offices, laboratories, or institutional environments. The robot will move autonomously along a fixed route, safely respond to obstacles, and deliver items to the intended person or location. The main goal is to reduce the time and effort required for repetitive indoor delivery tasks and make everyday service activities more convenient, organized, and efficient.

PROBLEM STATEMENT

- **Time Inefficiency:** Office staff and faculty spend valuable hours manually transferring physical papers and files between desks.
- **Lack of Confidentiality:** Sensitive documents, exam papers, or confidential files left on unattended desks pose security risks
- **Frequent Interruptions:** Manual deliveries disrupt focused work hours in quiet office environments.

OUR SOLUTION

- An autonomous, line-guided delivery robot equipped with real-time obstacle avoidance.
- Guarantees secure, trackable, and contactless intra-office document transportation.

KEY FEATURES

01 **Autonomous Line Following:** Follows designated indoor floor paths

02 **Turn-Sequence Navigation:** Manages multi-branch junctions and routes smoothly.

03 **Real-time Obstacle Avoidance:** Automatic stopping and turning upon sensing moving people or objects

KEY BENEFIT

Accurate, confidential delivery with zero manual carrying of files or sensitive documents.

- **Academic Campus:** Office-to-faculty room exam paper & file delivery.
- **Corporate Office:** Inter-departmental confidential document transport.
- **Research Labs:** Safe transport of small components, hard drives, and gear.
- **Administrative Wings:** Centralized mailroom-to-desk dispatch.

HARDWARE LAYER

The physical hardware framework that senses, moves, secures, and communicates.

SYSTEM COMPONENTS

- **ESP32 Development Board (CP2102, 38 Pin):** Main controller with Wi-Fi & processing capabilities.
- **RC522 RFID Reader Module:** Detects floor RFID tags and scans user cards.
- **RFID Tags & Cards:** Desk identification tags, colleague personal access cards, and junction/stop markers.
- **IR Sensor Array (3–5 Sensors):** Line-following track navigation.
- **Ultrasonic Sensor (HC-SR04):** Real-time obstacle detection.
- **Motor Driver (L298N) + 4x DC Gear Motors:** Differential drive propulsion.
- **Servo Motor:** Automated payload compartment lock/unlock mechanism.
- **Buzzer & LED Indicators:** Audio-visual arrival and obstacle alert signals.
- **Chassis, Wheels & Battery (7.4V / 12V Pack):** Physical structure and power distribution.

DECISION LAYER

Processor

ESP32 Development Board (CP2102, 38 Pin)

RESPONSIBILITIES

- **Path Tracking:** Reads 3–5 IR array sensors and calculates differential motor speeds via L298N driver for accurate line navigation.
- **Safety Assessment:** Continuously monitors HC-SR04 ultrasonic sensor data and halts movement instantly if an obstacle is detected within threshold distance.
- **Junction & Desk Recognition:** Reads floor RFID tags via the RC522 module to identify routing junctions and target desk stop locations.
- **Access Control Logic:** Validates scanned personal RFID cards against authorized IDs to actuate the Servo Motor for unlocking the compartment.
- **Audio-Visual Alerting:** Activates Buzzer beeps and LED indicators upon destination arrival or emergency obstacle stops.
- **Wireless Logging:** Sends real-time delivery status and arrival logs over Wi-Fi.

SYSTEM FLOW

INPUT LAYER

- IR Array (3–5): Line tracking data.
- Ultrasonic (HC-SR04): Obstacle distance.
- RC522 RFID: Floor tag & user card scan.
- ESP32 Wi-Fi: Wireless call request.

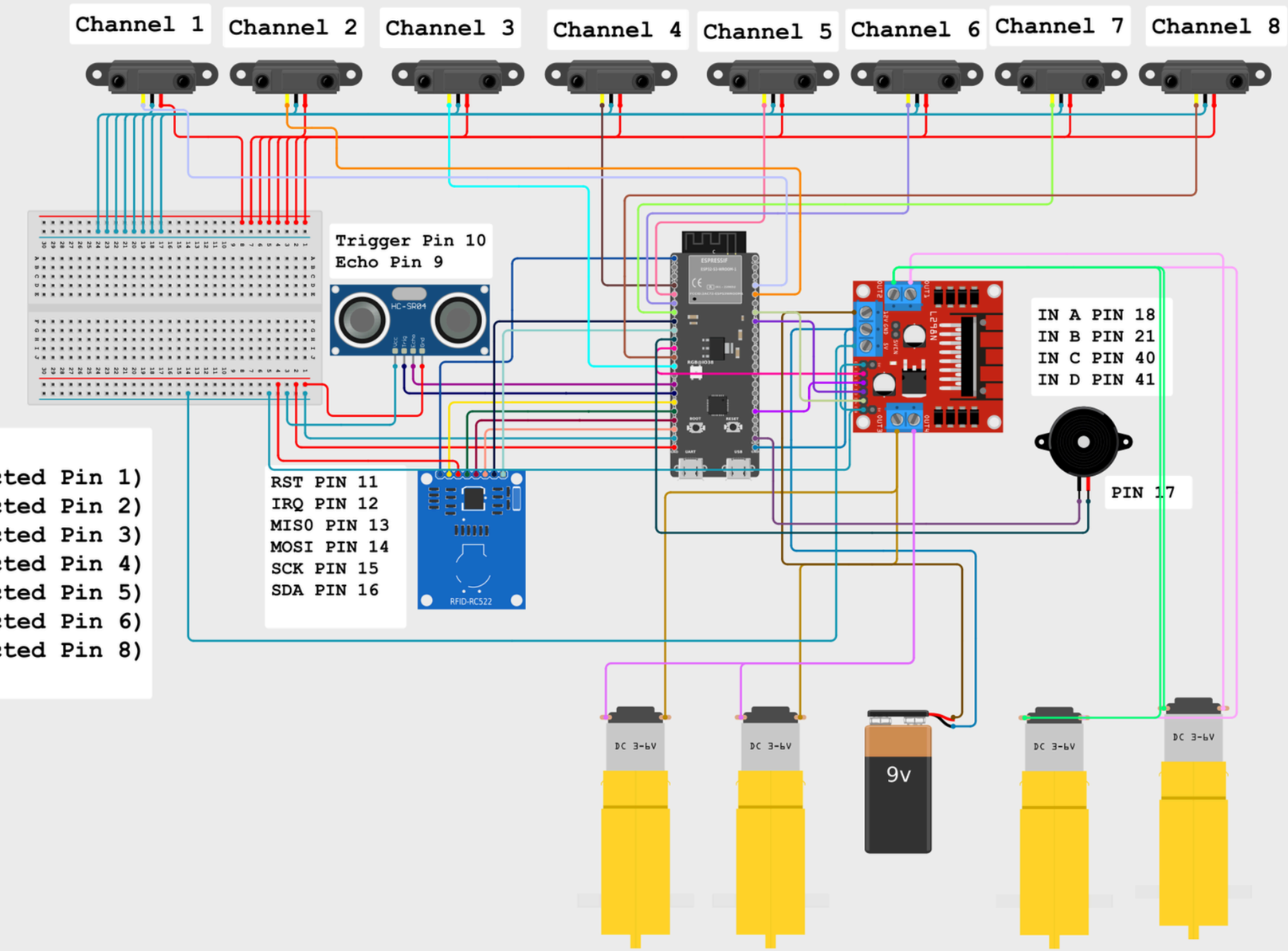
PROCESSING LAYER (ESP32)

- Calculates motor steering for line tracking.
- Halts system if obstacle is within range.
- Verifies desk RFID tags and authorized user cards.

OUTPUT

- L298N + DC Motors: Navigation & movement. .
- Buzzer : Arrival & obstacle alerts.

Circuit Design



For IR Sensor
Channel 1 (Connected Pin 1)
Channel 2 (Connected Pin 2)
Channel 3 (Connected Pin 3)
Channel 4 (Connected Pin 4)
Channel 5 (Connected Pin 5)
Channel 6 (Connected Pin 6)
Channel 8 (Connected Pin 8)

Sensor channel	ESP32 GPIO
Channel 1	GPIO34
Channel 2	GPIO35
Channel 3	GPIO36
Channel 4	GPIO39
Channel 5	GPIO13
Channel 6	GPIO14
Channel 7	GPIO16
Channel 8	GPIO17

Ultrasonic Trigger: GPIO27
Ultrasonic Echo: GPIO33
L298N IN1: GPIO25
L298N IN2: GPIO26
L298N IN3: GPIO32
L298N IN4: GPIO4
SDA/SS → GPIO5
SCK → GPIO18
MISO → GPIO19
MOSI → GPIO23
RST → GPIO22

HC - SR04

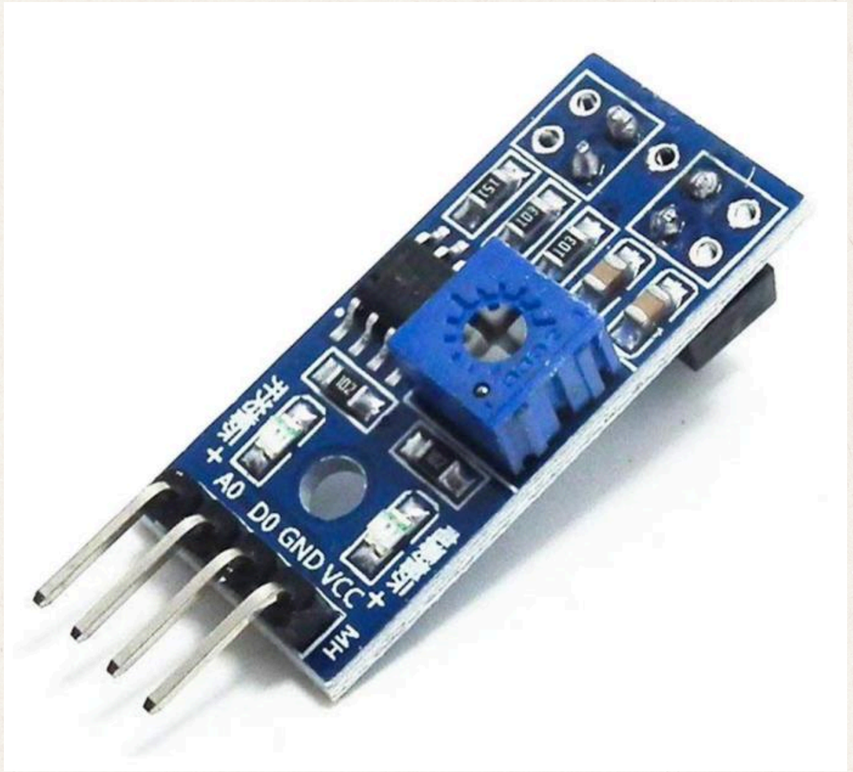
Pseudocode:

BEGIN PROGRAM
Configure TRIG_PIN as OUTPUT
Configure ECHO_PIN as INPUT
WHILE system_is_active DO:
 Write TRIG_PIN to LOW
 Wait for 2 microseconds
 Write TRIG_PIN to HIGH
 Wait for exactly 10 microseconds
 Write TRIG_PIN to LOW
 Record pulse duration from ECHO_PIN
 Current_Distance = Convert duration to centimeters
 IF Current_Distance < Minimum_Safe_Distance THEN
 Halt all motors
 Sound the alarm buzzer
 ELSE
 Drive motors forward
 Silence the alarm buzzer
 END IF
END WHILE
END PROGRAM

TCRT 5000

Pseudocode:

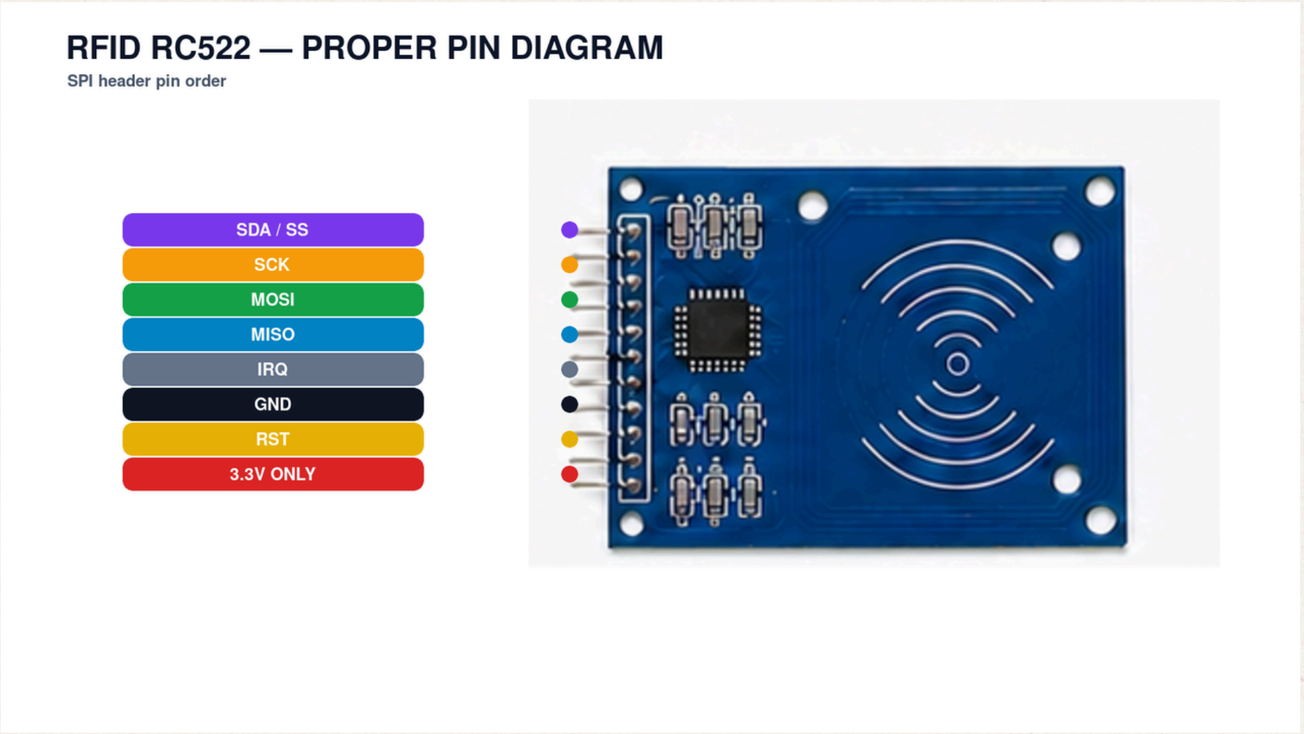
START
Initialize all 8 IR sensor pins as INPUT
Initialize motor control pins as OUTPUT
WHILE robot is ON:
 Read values from all 8 IR sensors (Sensors 1 through 8)
 IF center sensors (e.g., 4 and 5) detect the line THEN
 Move robot forward
 ELSE IF left-side sensors (e.g., 1, 2, or 3) detect the line
THEN
 Turn robot left
 ELSE IF right-side sensors (e.g., 6, 7, or 8) detect the line
THEN
 Turn robot right
 ELSE IF all 8 sensors detect the line THEN
 Stop robot
 ELSE
 Adjust direction to find the line
 END IF
END WHILE
END



RFID MFRC522

Pseudocode

```
BEGIN SYSTEM

Initialize RFID_Scanner ( MFRC522 )

WHILE System_Active == TRUE DO:

    IF Card_Present() == TRUE THEN

        Current_Tag_UID = Read_Card_Serial()

        IF Current_Tag_UID is in Authorized_Database THEN

            Trigger_Access_Granted()

        ELSE

            Trigger_Access_Denied()

        END IF

    END IF

END WHILE
END SYSTEM
```



Budget Estimation

Components	Quantity	Unit Price
IR sensor array	1	600
ESP32	1	620
Ultrasonic sensor(HC-SR04)	1	100
L298N motor driver	1	200
DC gear motor + wheel+Chassis	1	900
Battery (Li-ion/18650 pack)+Holder	3+1	400
RFID Reader module (RC522)	1	175
RFID tags/cards	1	50
Buck Converter	1	300
Jumper Wire	50	150
	Total	3,495

Contribution

Members	Contribution
Md. Osman Goni - 23101012	Hardware implementation; stakeholder management; survey administration; project testing; presentation preparation
Arannamoy Mondal - 23101024	Circuit development; programming; survey administration; project testing; project testing; presentation preparation
Fabia Akter Borsha - 23101025	Technical report preparation; survey administration; project testing; presentation preparation

Simulation

Click Here

Stakeholder Meeting

Current Office Logistics & Challenges

3. What is your role? *
- ☐ Executive
- ☐ Manager
- ☑ Administrative Staff
- ☐ Department Head
- ☐ Other: _____

4. How often do you send or receive physical items (files, documents, hard drives, mail) between offices/floors daily? *
- ☑ Multiple times a day (5+ times)
- ☐ Moderately (2–4 times a day)
- ☐ Rarely (Once a day or less)

5. How are documents and items currently delivered within your office? *
- ☑ Self-delivery (Walking manually between desks/floors)
- ☑ Dedicated office messenger / Assistant
- ☐ Left on desks or central mailroom until picked up

6. What are the main challenges with the current delivery process in your office? *
- ☑ Time lost walking back and forth during working hours
- ☑ Delays due to unavailability of office support staff
- ☑ Lack of real-time tracking (uncertainty if item reached recipient)
- ☑ Confidentiality/Security risks (important files left unattended on desks)
- ☑ Frequent interruptions during focus work hours

7. On average, how much time do you spend daily moving physical items within the office? *
- ☐ Less than 15 minutes
- ☐ 15 to 30 minutes
- ☑ 30 to 60 minutes
- ☑ More than 1 hour

11. What safety and communication features are most essential for office floor navigation? *
- ☑ Instant stopping upon sensing obstacles or moving people
- ☑ Subtle sound/buzzer alerts while moving
- ☐ Visual status indicators (LED signals)
- ☐ Designated floor track/lane markings

System Acceptance & Feedback

12. If deployed in your office, how likely are you to use this system for daily document transfers? (1 = Unlikely, 5 = Highly Likely) *
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☑ 5

13. What additional features or security measures would you recommend for corporate deployment?

_____
_____

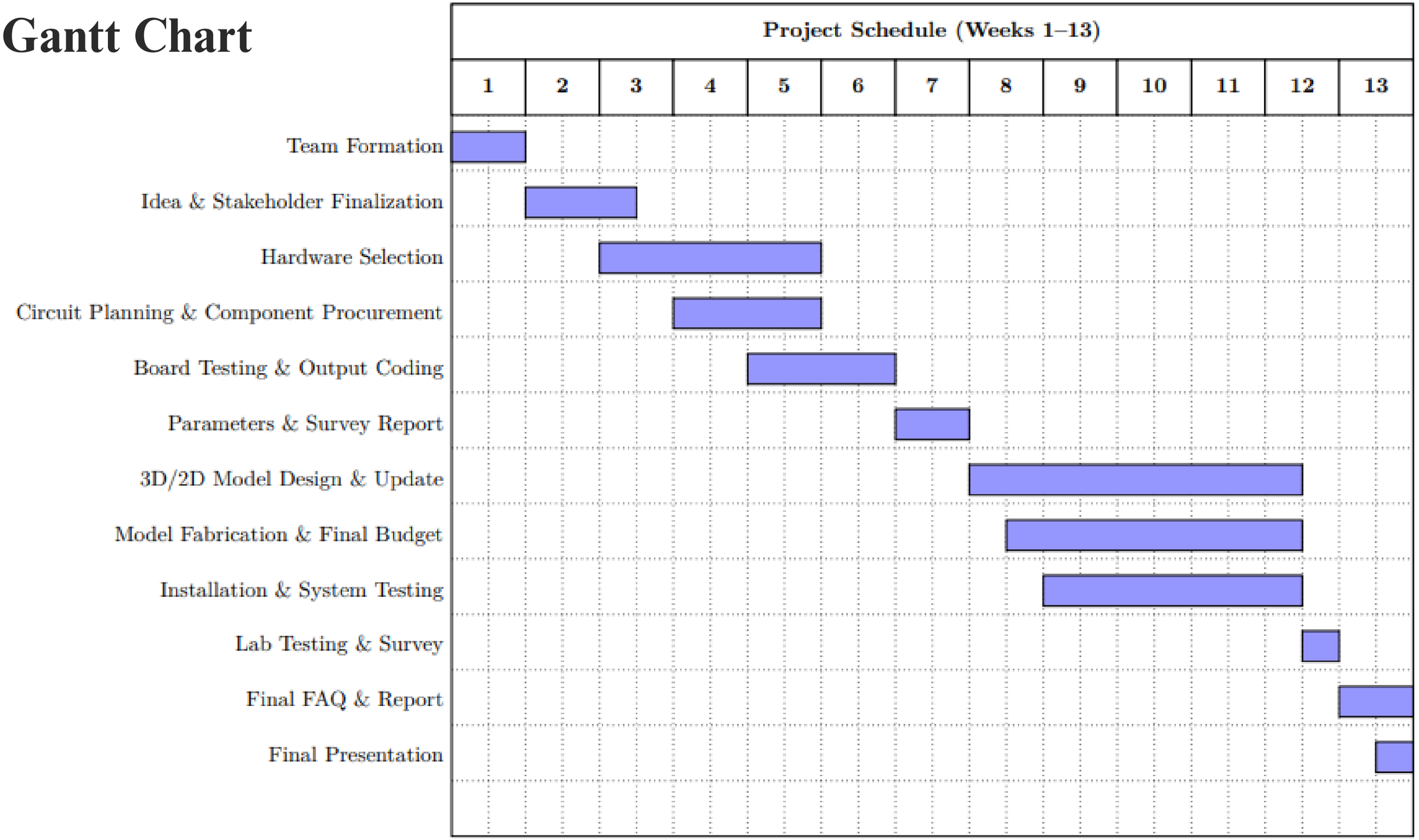
7.9.26

Stakeholder Signature

SETTING MEASURABLE PARAMETERS

No.	Measurable Parameter	Target / Setting	How It Is Measured
1	Line-Following Precision	Maximum deviation of +-1.5 cm from the designated path	Measure how far the robot moves away from the line using the IR sensors.
2	Obstacle Detection & Emergency Stop	Stop at 20cm; response time <=500ms	Use the HC-SR04 ultrasonic sensor to measure the obstacle distance and the time taken to stop the robot.
3	RFID-Based Stop Accuracy	Robot stops when the correct RFID tag is detected	Check whether the robot stops correctly when the assigned RFID tag is detected.

Gantt Chart


Project Schedule (Weeks 1–13)
1
2
3
4
5
6
7
8
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11
12
13
Team Formation
Idea & Stakeholder Finalization
Hardware Selection
Circuit Planning & Component Procurement
Board Testing & Output Coding
Parameters & Survey Report
3D/2D Model Design & Update
Model Fabrication & Final Budget
Installation & System Testing
Lab Testing & Survey
Final FAQ & Report
Final Presentation

Initial FAQ

- **Q01: What is this project?** An autonomous indoor robot that follows floor lines to deliver office documents to specific desks.
- **Q02: Is it fully autonomous?** Yes, line navigation, junction turns, obstacle avoidance, and target stopping are fully automated.
- **Q03: How is the delivery item delivered?** The robot carries the food or document on its tray and stops at the destination when the correct RFID card is detected.
- **Q04: What are the main hardware components?** ESP32, RC522 RFID module, IR sensor array, HC-SR04 ultrasonic sensor, L298N driver, DC gear motors, and Buzzer.
- **Q05: Why choose ESP32 over Arduino UNO?** Dual-core high processing speed, extra GPIOs, and built-in Wi-Fi/Bluetooth for wireless calls and status logging.
- **Q06: How does it handle turns at junctions?** The RC522 reader scans floor-embedded RFID tags at junctions to decide whether to turn or move straight.
- **Q07: How does the obstacle safety system work?** The HC-SR04 sensor measures distance ahead and halts the motors immediately if an object is within the safety threshold.
- **Q08: How does the robot identify the target desk?** It scans unique floor RFID tags at each station and stops upon matching the destination ID.
- **Q09: What happens if the robot passes over a different RFID card?** If the RFID does not match the destination ID, the robot ignores it and continues moving.
- **Q10: What are the main technical success metrics?** +-1.5cm line accuracy, <=500ms obstacle brake response, and 100% RFID verification accuracy.
- **Q11: Who are the primary stakeholders and target users?** University office staff, corporate executives, administrative faculty, and project evaluators.

SURVEY#1 REPORTING

1. Operational Logistics

- **Respondent Role:** Administrative Staff / Office Personnel / Campus Coordinator
- **Manual Delivery Frequency:** Multiple times per hour during peak operations
- **Time Spent Per Shift:** Significant working hours spent carrying trays/items manually
- **Most Common Transport Items:** Food and physical files
- **Primary Expected Benefit:** Reduce manual human effort and enable faster, safer serving

2. System Preferences

- **Navigation & Route:** Line-following with designated routes for each table/desk
- **Hallway & Floor Conditions:** Dynamic indoor spaces — real-time obstacle avoidance essential
- **Delivery Compartment:** Stable tray carrying
- **User Control Interface:** Predefined destination selection & automated line tracking
- **Status & Warning Alerts:** Buzzer sounds for arrival and alerts

3. Priorities & Acceptance

- **Top Design Priorities:** Accurate table-wise delivery | Autonomous line precision | Safe operation
- **Key Concerns to Address:** Obstacle collision | Food spillages / tray balance | Path navigation errors
- **Expected Reliability Threshold:** High operational uptime during service hours
- **Final Acceptance Decision:** Approved for smart indoor deployment in office or workspaces